

I fuse machine learning with numerical methods to accelerate materials discovery and design. I design piecewise-deterministic Markov process (PDMP) samplers with superior mixing, and pair them with surrogate models so they scale to problems with expensive forward simulations. This makes uncertainty quantification fast and reliable, enabling principled reasoning and well-founded decisions.

## Education

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| <b>Doctor of Philosophy</b><br><i>Civil Engineering</i>                                  | <b>Delft University of Technology, NL</b><br>07/2021 – present |
| <b>Thesis:</b> Efficient Bayesian Inference of Parametric Partial Differential Equations |                                                                |
| <b>Master of Science</b><br><i>Mechanical Engineering</i>                                | <b>Technical University Munich, GER</b><br>03/2018 – 05/2021   |
| <b>Bachelor of Science</b><br><i>Mechanical Engineering</i>                              | <b>Technical University Munich, GER</b><br>10/2013 – 02/2018   |
| <i>Semesters abroad:</i>                                                                 |                                                                |
| <b>Keio University</b> , Tokyo, JP                                                       | 03/2019 – 09/2019                                              |
| <b>Ecole Polytechnique de Montréal</b> , Montréal, CA                                    | 01/2016 – 05/2016                                              |

## Research

### Selected publications

- [1] Leon Riccius, Atul Agrawal, and Phaedon-Stelios Koutsourelakis. Physics-Informed Tensor Basis Neural Network for Turbulence Closure Modeling. In *37th Conference on Neural Information Processing Systems, NeurIPS*, 2023.
- [2] Leon Riccius, Iuri Rocha, Joris Bierkens, Hanne Kekkonen, and Frans van der Meer. Integration of Active Learning and MCMC Sampling for Efficient Bayesian Calibration of Mechanical Properties. *European Journal of Mechanics A/Solids*, 2026.
- [3] Leon Riccius, Iuri Rocha, Joris Bierkens, Hanne Kekkonen, and Frans van der Meer. Piecewise Deterministic Markov Processes for Bayesian Inference of PDE Coefficients. 2026. under review.

I presented my work at 7 international conferences, including SIAM AN, ECCOMAS, and UNCE-COMP.

## Machine Learning Experience

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|------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Modeling:</b>             | Gaussian Process, Neural Networks, Variational Autoencoders (VAE), Conditional VAEs, Diffusion Models |
| <b>Software &amp; Tools:</b> | Python (pytorch, jax), C++, bash, git                                                                 |
| <b>Communication:</b>        | Data visualization, academic writing, public speaking, supervision                                    |

## Github Repositories

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[SLIMM-Lab/pdmp](#): Library for approximating complex probability distributions via piecewise deterministic Markov process sampling (pytorch based ML, jax based differentiable FEM models)  
[SLIMM-Lab/mcmc-with-surrogates](#): Bayesian inference of FEM model parameters where expensive likelihoods are approximated with Gaussian process models (C++ based)

## Awards

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| <b>Travel Award</b><br><i>Society for Industrial and Applied Mathematics (SIAM)</i>       | <b>Montréal, CA</b><br>2025 |
| <b>Best Poster Award</b><br><i>27th Engineering Mechanics Symposium (114 submissions)</i> | <b>Papendal, NL</b><br>2024 |

## Teaching

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| <b>TU Delft, Faculty for Civil Engineering and Geoscience</b><br><i>Designing <a href="#">interactive textbook</a><sup>1</sup>, teaching assistant<sup>2</sup>, project supervisor<sup>3</sup></i> <ul style="list-style-type: none"><li>◦ Modeling, Uncertainty, and Data Analysis for Engineers<sup>1,2,3</sup></li><li>◦ Data Science and Artificial Intelligence for Engineers<sup>1,2,3</sup></li><li>◦ Statistical Learning for Engineers<sup>2</sup></li></ul> | <b>Delft, NL</b><br>07/2021 – present |
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### Thesis Supervision:

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| Daan Smolders, Master Thesis, in cooperation with Airbus                                                     | 2024 |
| <i>Surrogate modelling for uncertainty propagation by targeting regions of high variance and probability</i> |      |

## Languages

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|------------------|-------------------------------|-----------------------------|
| <b>German:</b>   | Native proficiency            |                             |
| <b>English:</b>  | Full professional proficiency | <i>TOEFL iBT Score: 111</i> |
| <b>French:</b>   | Limited working proficiency   | <i>Abitur: B2/C1</i>        |
| <b>Dutch:</b>    | Limited working proficiency   |                             |
| <b>Japanese:</b> | Elementary proficiency        |                             |